

# The Phonology of Indonesian

**Author:** Dan Brodtkin

**Affiliation:** University of California, Santa Cruz

**Email:** ddbrodki@ucsc.edu

**Abstract:** This paper reviews the phonology of Standard Indonesian, focusing on the following: (i) the segmental inventory, (ii) the segments /ʔ h ə/, (iii) syllable structure, (iv) positional constraints on segments, (v) nasal substitution, (vi) reduplication, (vii) word-level stress, (viii) foot structure, and (ix) prosodic phrasing. Comparisons are made with the most important languages in the Indonesian linguistic ecosystem (Javanese, Jakarta Indonesian); the paper concludes with a reflection on the emerging direction of work on the controversial topic of word-level stress.

**Keywords:**

Foot Structure, Glottal Segments, Indonesian, Nasal Substitution, Prosody, Reduplication, Stress

**Key Points:**

- Description of the segmental inventory; restrictions on segments; phonology of /ʔ h ə /
- Discussion of nasal substitution (Pater 1999); connection to parallel research on Malay
- Discussion of word-level prominence; evidence for metrical structure; prosodic phrasing

**Introduction:** Bahasa Indonesia, or *Indonesian*, is the national language of Indonesia. It is a Malayic language of the Malayo-Polynesian subgroup of Austronesian and is closely related to the standardized varieties of Malay official in Malaysia, Singapore, and Brunei. Standard Indonesian originates most directly from the literary Malay of the courts of Malacca, Johore, and Riau, which was selected over many forms of “low Malay” and cultivated as the main language of colonial administration by the Dutch (Adelaar, 2000). This language was cemented as the primary vehicle of the nationalist movement in 1928 and has since risen to become the sole language of government and education across the country, where it is now spoken natively by around 71,000,000 people and spoken as a second language by 176,500,000 others (Badan Pusat Statistik Indonesia, 2020).

Standard Indonesian is an acrolectal language that must be studied and discussed against a complex linguistic backdrop. There are over 700 regional languages still living in Indonesia today, and of these the standard language interacts in an especially tight fashion with (a) regional lingua francas that are lexified primarily by Indonesian but show immense variation in all other ways, (b) Jakarta Indonesian, the lingua franca that has emerged around the capital (Wallace, 1976), and (c) Javanese, the regional language with the greatest number of native speakers, highest prestige, and widest geographic distribution, and the distinctly Javanese variety of Indonesian built upon it. Without exception, speakers of Standard Indonesian have command over other linguistic systems of this type, in addition to the regional languages they may already speak, and even those who are not native speakers of Javanese Indonesian or Jakarta Indonesian are often well-exposed to both. For this reason, speakers are likely to draw on different and varying linguistic subsystems when asked to produce Standard Indonesian, and as a result, there is disagreement in the literature over many properties of the language. The goal of this review, then, is to organize and present the generalizations most widely accepted of the standard language, reinforce them with parallels to Jakarta Indonesian and Javanese, and highlight the remaining descriptive challenges that persist.

**Consonants:** Table (1) presents the consonantal phoneme inventory of Standard Indonesian, using parentheses for the phonemes that appear only in the non-native strata of the lexicon. The language contrasts three series of stops and a series of palatal affricates, along with four nasals at the corresponding points of articulation; the dental /t n/ also differ in place from the postalveolar /d r/ (as Blust 2013 claims to be common across Austronesian). The glides /w j/ are phonemic, though in proto-Malayic they were not (Adelaar, 1985). The situation with fricatives is complex: many early borrowings whose source forms contain /f ɕ z x/ come down to the standard language as doublets, showing the alternations *f-p* (*hafal/hapal* “memorize”), *ɕ-s* (*syiwa/siwa* “Shiva”), *z-dz/s* (*zaman/jaman* “era”), *x-k/h* (*khawatir/kuatir* “worry”). Alieva et al. 1972 thus argue that /f ɕ z x/ are not phonemic, but it seems clear that they now are: many borrowings are now exclusively realized with /f ɕ z x/, there are many minimal pairs for *p/f* (*fakta* ‘fact’ vs. *pakta* ‘pact’), and English loans are increasingly adapted with unassimilated /z ɕ (<ʃ)/ (Stokhof 1975; Mahdi 1981).

(1) Table One: Consonant Phonemes of Standard Indonesian

|           | Bilabial | Labiodental | Dental | Postalveolar | Palatal | Velar | Glottal |
|-----------|----------|-------------|--------|--------------|---------|-------|---------|
| STOP      | p b      |             | t d    |              |         | k g   | ʔ       |
| AFFRICATE |          |             |        |              | tɕ dʒ   |       |         |
| NASAL     | m        |             | n      |              | ɲ       | ŋ     |         |
| FRICATIVE |          | (f)         | s (z)  |              | (ç)     | (x)   | h       |
| LIQUID    |          |             | l      |              |         |       |         |
| RHOTIC    |          |             |        | r            |         |       |         |
| GLIDE     | w        |             |        |              | j       |       |         |

**Glottals:** Standard Indonesian has two glottals, /h ʔ/, of which only /h/ is universally termed phonemic. /h/ surfaces not only in onset positions but also in syllable codas, both word-finally (in the native stratum: *basah* ‘wet’) and word-medially (mainly in Arabic loans: *tahta* ‘throne’). **Lapoliwa 1981** describes several patterns of /h/-deletion that seem, to my judgment, to be sensitive to the positions of words in higher prosodic domains: word-final /h/ is typically deleted; root-final /h/ is deleted before suffixes (thus /*kə-lətih-an*/ ‘tiredness’ > [*kələtian*]), unless the result would be hiatus of identical vowels (thus /*kə-basah-an*/ ‘wetness’ > [*kəbasahan*]), and intervocalic /h/ is deleted between high-low and low-high sequences (/lihat/ ‘see’ > [*liat*]; /*d̥zahit*/ ‘sew’ > [*d̥zait*]; cf. /*sehat*/ ‘healthy’ > [*sehat*]) unless the root-final segment is a liquid (/lahir/ ‘born’ > [*lahir*]). **Lapoliwa 1981** claims further that the process is category-sensitive, failing to occur in nouns (thus /*tahu*/ ‘know’ (v) > [*tau*] but /*tahu*/ ‘tofu’ (N) > [*tahu*]), though he cites several counterexamples.

The status of /ʔ/ is more complex. This segment is consistently epenthesized at the left edges of vowel-initial prosodic words; **Mahdi 1981** notes that it prevents resyllabification in compounds (/tanah-air/ ‘motherland’ > [*tana ʔair*]); it also reveals the presence of prosodic word boundaries after certain prefixes (/bər-efolusi/ ‘evolve’ > [bərʔefolusi]; *di-ambil* ‘taken’ > [*diʔambil*]; see also **Cohn & McCarthy 1998**). The glottal stop also appears between sequences of like vowels (*sāʔāt* ‘time’) and mid-low sequences (*soʔal* ‘matter’) in borrowings from Arabic, corresponding to /ʔ ʔ/; it is also inserted to break up identical vowel sequences after the nominalizing suffix -an (/bat̤ca-an/ ‘reading’ > [bat̤caʔan] but /tari-an/ ‘dance’ > [tarijan]). In all intervocalic positions it is liable to deletion, though this pattern is also likely conditioned by prosodic phrasing.

The Arabic loan stratum contains a number of roots that must contain underlying /ʔ/, such as /*d̥zumʔat*/ ‘Friday’ and /*kurʔan*/ ‘Qur’an’. **Mahdi 1981** notes that there are also minimal pairs in the native stratum that contrast /ʔ/ with /k/ in word-final positions (where /p t k/ are typically unreleased): /*paʔ*/ ‘father, sir’ – /*pak*/ ‘pack, /*masaʔ*/ ‘no way’ – /*masak*/ ‘cook’ (see also **Tadmor 2007**, 304:305). These facts suggest that /ʔ/ must be a phoneme (cf. **Stokhof 1975**; **Lapoliwa 1981**).

**Vowels:** Table (2) presents the vowel inventory, using parentheses for the phonemes that appear only in the non-native strata. Standard Indonesian has six native vowels: /i e a ɔ u ə/ (**Alieva et al.**, 1972; **Lapoliwa**, 1981). The mid-open /e ɔ/ are innovations from Proto-Malayic (**Adelaar**, 1985) that contrast with /i u/, though doublets persist (*senin/senen* ‘monday’, *lubang/lobang* ‘hole’). **Lapoliwa 1981** claims that these vowels are underlying /e o/ and argues that they lower to /e ɔ/ in closed final syllables, but the situation is more complex: /e o/ contrast with /e ɔ/ in Javanese, and this contrast persists so regularly in borrowings from that language that **Samsuri 1960** and **Dardjowidjojo 1966** propose that /e o/ contrast with /e ɔ/ in Standard Indonesian as well. In support of this view, **Mahdi 1981** notes that /e o e ɔ/ have stable and contrastive distributions in loanwords from English and Dutch, adducing minimal pairs like /*kopi*/ ‘copy’ – /*kɔpi*/ ‘coffee’. He also notes a contrast between /i/ and /ɪ/, as in /*d̥zɪp*/ ‘jib’ – /*d̥zɪp*/ ‘jeep’, /*nasi tɪm*/ ‘rice broth’ – /*tɪm*/ ‘team.’

(2) Table Two: Vowel Phonemes of Standard Indonesian

|           | Front | Mid | Back |
|-----------|-------|-----|------|
| HIGH      | i (ɪ) |     | u    |
| MID-CLOSE | (e)   |     | (o)  |
| MID-OPEN  | ɛ     | ə   | ɔ    |
| LOW       |       | a   |      |

**Schwa:** There are a number of prosodic regularities that surround the distribution of schwa. That segment is banned from word-final syllables in the native vocabulary, though it emerges

in that position in loans from Javanese and Dutch (/haltə bəs/ ‘bus stop). In antepenultimate positions, the language historically reduced all vowels to /ə/ (Adelaar, 1985); Lapoliwa 1981 notes that a similar process seems to operate in the synchronic phonology. Grijns 1981 and Cohn 1989 propose fully derivational accounts of the distribution of /ə/, focusing on patterns of epenthesis that operate in the loan strata from English and Dutch. Adisasmito 1993, however, shows that /ə/ must be posited as an underlying phoneme in at least some native and borrowed roots. On the view that there are underlying cases of /ə/, it is then necessary to posit a process of /ə/-deletion to capture regular /ə/–/Ø/ alternations, like /bəli/ ‘buy’ > [bəli] but /bəli ini/ ‘buy this’ > [bli ini].

**Nasalization:** Nasalization is not contrastive in Standard Indonesian but emerges in two contexts. First, nasal consonants trigger nasalization of the immediately following vowel (extending farther in related languages (Court, 1970); for formal discussion, Walker 1998; McCarthy 2004). Second, Court 1970 and Lapoliwa 1981 claim that nasalization emerges in borrowings from Arabic that contain like vowels separated by /ʔ/ (etymologically /ʕ/): thus *sāʔāt* ‘time,’ *tāʔāt* ‘faithful.’

**Syllable Structure:** the maximally articulated syllable structure within the native vocabulary was at least historically (c)v(c); within the various loan strata it now reaches up to (c)(c)(c)v(c)(c). The language historically restricted cc clusters to word-medial positions and allowed only rc, /ŋs/, and homorganic nasal-obstruent sequences, reducing all other types of inherited medial clusters. The same treatment was extended to early loanwords from Arabic (/mumkin/ ‘possible’ > munkin), but the language has now acquired many more types of medial clusters, including heterorganic sequences of (non-palatal) nasal + stop (*jamrud* ‘emerald,’ *tanpa* ‘without,’ *anpao* ‘gift envelope’). The language allows stop-liquid onset clusters in loans from English, Dutch, and Javanese (*kredit* ‘credit,’ *pretel* ‘dismantle’) and in loans from English and Dutch, it also allows onset clusters of sibilant and stop (*stasiun* ‘station,’ *skala* ‘scale’) and sibilant, stop, and liquid (*skripsi* ‘undergraduate thesis’); for many speakers these clusters are never split by /ə/-epenthesis (Adisasmito, 1993). Word-finally, in loans from English and Dutch, stop-sibilant clusters are increasingly retained intact (*kompleks* ‘complex,’ *indeks* ‘index’); at least once it was typical for liquid-stop clusters to be reduced (*sport* > *spor*) and liquid-nasal clusters split by /ɛ/-epenthesis (*modern* > *mōdɛrən*–*mōdɛrən*), but the trend in each case is now toward complete and unassimilated retention (Adisasmito, 1993).

**Positional Restrictions on Segments:** Standard Indonesian prohibits syllable-final /t͡ɕ d͡ʒ/. There is only one context in which the syllable-final position seems to tolerate the palatal nasal /ɲ/: in word-medial sequences of /pt͡ɕ/ and /ɲd͡ʒ/ (e.g., *ap̚t͡ɕaŋ* ‘threaten’). But this apparent exception may be illusory if homorganic nc-sequences are parsed as complex onsets, as they are in Javanese. Standard Indonesian is traditionally held to ban word-final voiced obstruents (Lapoliwa, 1981), yielding regular alternations in the loan strata (/d͡ʒawab/ ‘respond’ > [d͡ʒawap] but /d͡ʒawab-an/ ‘response’ > [d͡ʒawaban]); this ban is absent from English and Javanese and seems increasingly unenforced in the strata of loans from those languages (thus /gud͡ʒg/ ‘jackfruit dish’ > [gud͡ʒg]). There are several further restrictions on vowel cooccurrence in the native stratum. Lapoliwa 1981 notes two: the penultimate syllable never hosts the mid-vowels (his /e o/) when the final syllable hosts /i u/ and never hosts /u/ when the final syllable contains his /o/. Mahdi 1981 notes further, despite a handful of exceptions, that the mid-close /e o/ rarely occur in the same words as /ɛ ɔ/.

**Nasal Substitution:** Standard Indonesian has two prefixes, the active voice prefix *meN-* and the nominalizer *peN-*, whose final segment triggers a special process: before vowel-initial roots it surfaces as [ɲ] (/məN-alir/ ‘flow’ > [məɲalir]), before voiced obstruents it assimilates in place (3a)-(3c), and before voiceless stops it assimilates and fuses with the following stop (3d)-(3f).

### (3) Nasal Substitution

- |                                               |                                                  |
|-----------------------------------------------|--------------------------------------------------|
| a. /mən-bat̪ɕa/ ‘read’ > [mə <b>m</b> bat̪ɕa] | d. /mən-pətɔŋ/ ‘cut’ > [mə <b>m</b> ɔtɔŋ]        |
| b. /mən-dɔrɔŋ/ ‘push’ > [mə <b>n</b> dɔrɔŋ]   | e. /mən-tɔlɔŋ/ ‘help’ > [mə <b>n</b> ɔlɔŋ]       |
| c. /mən-gali/ ‘dig’ > [mə <b>ŋ</b> gali]      | f. /mən-kandun/ ‘contain’ > [mə <b>ŋ</b> kandun] |

This process was already established in Proto-Malayo-Polynesian (Blust, 2004), and it surfaces in similar ways in Jakarta Indonesian and Javanese (Kurniawan, 2018). In Standard Indonesian, as in those languages, the process substitutes /s/ with /ɲ/ (/mən-sərap/ ‘absorb’ > [məɲərap]), though in specific loanwords where substitution fails before /s/, the segment assimilates to /n/ instead (/mən-supsi/ ‘subsidize’ > [mənsupsi]; on substitution in loans, Sneddon 1996, 9-12). In Standard Indonesian, it fails to apply before the palatal /t̪ɕ/ (/pən-t̪ɕut̪ɕi/ ‘washer’ > [pənt̪ɕut̪ɕi]), but the cognate process substitutes /t̪ɕ/ in both Jakarta Indonesian (thus [pəɲut̪ɕi]) and Javanese. For valuable formal discussion, see Pater 1996, 1999, 2001; Archangeli *et al.* 1998; Nomoto 2009.

Nasal substitution shows two further layers of complexity in the wider system of prefixation. First, the process fails to apply in the complex prefix *meN-per-* (where *per-* is a causative): thus /mən-pər-kət̪ɕil/ ‘make smaller’ > [mə**m**pər-kət̪ɕil]; the situation is more complex with different varieties of *pər-* and with *peN-* (Sneddon, 1996, 12). Second, the process does apply between *meN-* and the prefixes *se-* ‘one’ (/mən-sə-ragam-kan/ ‘standardize’ > [məɲəragamkan]) and *ke-* (/mən-kə-tawa-i/ ‘laugh at’ > [məɲətawai]). In Malay, Delilkan 2002 argues that the same asymmetries reflect differences in the internal prosodic organization of these two types of prefixal strings—and in particular, provide evidence for the existence of metrical footing and specific types of feet. The facts in Standard Indonesian, then, may inform future work on metrical structure in the language.

**Reduplication:** Indonesian shows three patterns of reduplication. The first and most productive pattern copies full prosodic words: thus /pər-əmpat-an-RED/ ‘four way intersections’ > [pərəmpatan-pərəmpatan]. When this process targets verbs, it applies beneath the voice prefixes, and thus *meN-*, *ber-*, and *di-* are never copied: thus /mən-əlu-RED-kan/ ‘praise repeatedly’ > [məɲəlu-əlukan]. When *meN-* triggers nasal substitution of the onset of the verbal root, in turn, substitution persists in the reduplicant: /mən-kata-RED-i/ ‘scold repeatedly’ > [məɲata-ɲatai] (Lapoliwa 1981; Cohn & McCarthy 1998). Beyond this pattern, a type of partial *cə*-reduplication persists in a set of fossilized forms: *jaka* ‘bachelor’—*jə-jaka* ‘bachelor’, *luhur* ‘noble’—*lə-luhur* ‘ancestors’, *daun* ‘leaf’—*də-daunan* ‘leaves’, *jamur* ‘mushroom’—*jə-jamuran* ‘mushrooms. Finally, there is a third pattern, parallel to what Abdullah 1974 calls “Rhyming and Chiming Reduplication” in Malay, in which prosodic words are reduplicated and subjected to phonological change. In some cases, this targets the onset (*sayur* ‘vegetable’ > *sayur-mayur* ‘all sorts of vegetables’, *lauk* ‘side dish’ > *lauk-pauk* ‘all kinds of side dishes’); more often, the vowels change in regular ways (*balik* ‘return’ > *bolak-balik* ‘go back and forth’; *ganti* ‘change’ > *gonta-ganti* ‘change back and forth’). See Lapoliwa 1981, 189-192 for the main patterns and Soh 2011 for analysis in Malay.

**Phonologically Conditioned Allomorphy:** the voice prefixes *ber-* and *ter-* and the nominalizer *per-* take forms without the final /r/ before roots whose first syllables have the shape /cər/: /bər-kərd̪ʒa/ ‘work’ > [bəkərd̪ʒa], /bər-pərgi-an/ ‘go about’ > [bəpərgian]; /tər-pərgək/ ‘caught’ > [təpərgək]; /pər-tərnək-an/ ‘animal husbandry’ > [pətərnakan]. The effect also extends to certain roots where later syllables host coda /r/: /tər-lantar/ ‘abandoned’ > [təlantar] (Sneddon, 1996, 8).

**Stress:** Traditional sources present a relatively consistent picture of word-level stress: in each prosodic word, stress falls on the penult but shifts to the ultima if the nucleus of the penult is /ə/ (Alieva *et al.* 1972; Stokhof 1975; Lapoliwa 1981; Teeuw 1984; Cohn 1989; Cohn & McCarthy



1998; there is minor dispute on stress placement in polysyllabic words and the effects of certain affixes, but all sources agree that the placement of word-level stress interacts with a complex phrasal phonology that overrides the default patterns of footing at high-level prosodic junctures). An alternative view takes the interaction with higher-level prosody to suggest that there is no word-level stress (Halim, 1974; van Zanten & van Heuven, 1998; Goedemans & van Zanten, 2007), attributing the evidence traditionally marshalled in favor of word-level stress to interference from substrate languages and misperception by native listeners of languages that do have word stress

**Foot Structure:** Whatever the status of stress, there is clear evidence in Standard Indonesian for both word-internal metrical structure and for a systematic preference to build disyllabic feet. The language has a wide class of borrowed monosyllabic roots, but these invariably fail to combine with the prefixes *meN-/peN-* in the regular way. The attested pattern of affixation is shown in (4): when *meN-/peN-* select monosyllabic roots, they force the insertion of an intermediate prefix *kə-* (just as the passive prefix *di-* forces the insertion of *kə-* before the native monosyllabic root *tau* ‘know’: *di-kə-tau-i* ‘known’). The same pattern occurs in many further languages of the region, like Javanese (Wedhawati *et al.*, 2001), Sundanese (Robins, 1953), and Riau Indonesian (Gil, 2020).

(4) *Monosyllabic Roots + meN/peN-: Augmentation*

- |                                                        |                                                     |
|--------------------------------------------------------|-----------------------------------------------------|
| a. / <i>məN-pəl</i> / ‘mop’ > [ <i>məŋəpəl</i> ]       | d. / <i>məN-bər</i> / ‘drill’ > [ <i>məŋəbər</i> ]  |
| b. / <i>məN-tik</i> / ‘type’ > [ <i>məŋətik</i> ]      | e. / <i>məN-rəm</i> / ‘brake’ > [ <i>məŋərəm</i> ]  |
| c. / <i>məN-t̪ɛat</i> / ‘paint’ > [ <i>məŋət̪ɛat</i> ] | f. / <i>pəN-bəm</i> / ‘bomber’ > [ <i>pəŋəbəm</i> ] |

This is a minimality effect of the type most naturally understood by reference to foot structure (Itô & Mester, 1992): *meN-/peN-* force the insertion of /*kə*/ before monosyllabic roots because they force their complements to form binary feet (a case of prosodic subcategorization: Inkelas 1989). This pattern thus provides evidence for the existence of disyllabic feet in Standard Indonesian, and it converges with many more lines of evidence to show that such feet are systematically preferred:

(5) *Arguments for Word-Final Disyllabic Feet:*

- The vast majority of native roots are disyllabic;
- Antepenultimate vowels are historically and often synchronically reduced to /ə/, as if there is a process of reduction that is specifically blocked in word-final disyllabic feet;
- There is a process that reduces prevocalic high vowels to glides (/suatu/ ‘one’ > [swatu]), and it is blocked whenever it would create monosyllabic words (/suap/ ‘feed’ > [suwap], [\*swap]) (Lapoliwa, 1981, 185-187), as if it avoided the creation of monosyllabic feet;
- there is a productive process of blending that typically reduces multi-word strings to two-syllable words, as if its target were this foot (*budak cinta* ‘love slave’ > *bucin*; *malas gerak* ‘too lazy to move’ > *mager*; *menteri luar negri* ‘foreign minister’ > *menlu*);
- there are intonational contours that fall on a two-syllable window at the right edge of the intonational phrase, like the contradiction contour (*bi<sup>LH</sup> sa<sup>LH</sup>* “it is possible”); and
- there are language games that reveal the existence of disyllabic foot structure in Jakarta Indonesian (as well as in Riau Indonesian and Papuan Malay; Gil 2022).

**Prosodic Phrasing:** above the level of words and feet, Halim 1974 argues for the existence of two higher prosodic domains: restricting his attention to South Sumatra Indonesian, he proposes that words are parsed into *pause groups* and *pause groups* are parsed into *intonational patterns*.

These domains correspond neatly to the universal prosodic categories of *phonological phrase* ( $\phi$ ) and *intonational phrase* ( $\iota$ ) in the framework of Ito & Mester 2007 and Selkirk 2009, and the labels  $\phi/\iota$  are adopted for parallel categories in Ambon Malay by Maskikit-Essed & Gussenhoven 2016. There are many ways in which the segmental phonology of the language seems to be sensitive to the edges of these higher prosodic domains. In Jakarta Indonesian, for instance, Kurniawan 2018, 101 notes that word-final /h/ deletes only in words that are non-final in a large prosodic domain (in our terms, the  $\iota$ ). Tadmor 2007, 304, similarly, alludes to the fact that there are intonational tunes in Standard Indonesian that force epenthesis of /ʔ/ after vowel-final words at the right edge of the  $\iota$ . Impressionistically, there seem to be many further processes in Standard Indonesian that are conditioned by the positions of words within the  $\phi$ : for instance, there is a process of coalescence that typically reduces /mau/ ‘want’ to [mo] (6a) and /saja/ ‘me’ to [se] (6b), and this process is blocked whenever these words fall at the right edge of the  $\phi$  (6c)-(6d). It also seems likely that the  $\phi$  and  $\iota$  impose constraints on the shapes of the feet that fall at their right edges: specifically, the right edge of the  $\phi$  may impose a violable requirement that its final foot be disyllabic, as roots like /bəli/ can only undergo ə-deletion outside of  $\phi$ -final positions (7).

(6) *Vowel Coalescence: Conditioned by Position in the Phrase*

- |                                                                                                                                                                                     |                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a. <math>\{\phi</math> <b>mo</b> ini <math>\}</math> <math>\{\phi</math> tadi <math>\}</math><br/>           want this      earlier<br/>           ‘He wanted this earlier.’</p> | <p>c. <math>\{\phi</math> <b>mau</b> <math>\}</math> <math>\{\phi</math> tadi <math>\}</math><br/>           want      earlier<br/>           ‘He wanted (it) earlier.’</p>    |
| <p>b. <math>\{\phi</math> <b>se</b> suka <math>\}</math> <math>\{\phi</math> sih <math>\}</math><br/>           i like      actually<br/>           ‘I like (it), actually.’</p>    | <p>d. <math>\{\phi</math> kalo <b>saja</b> <math>\}</math> <math>\{\phi</math> ndak <math>\}</math><br/>           as for me      not<br/>           ‘As for me, I don’t.’</p> |

(7) *Phrase-Final Position: Stricter Requirements for Disyllabic Feet*

- |                                                                                                                                                                           |                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a. <math>\{\phi</math> <b>bli</b> ini <math>\}</math> <math>\{\phi</math> dulu <math>\}</math><br/>           buy this      first<br/>           ‘Buy this first!’</p> | <p>b. <math>\{\phi</math> <b>bəli</b> <math>\}</math> <math>\{\phi</math> dulu <math>\}</math><br/>           buy      first<br/>           ‘Buy (some) first!’</p> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Conclusion:** on the horizon, the deepest questions in the phonology of Standard Indonesian lie squarely in the domain of word-level stress, where researchers are beginning to disentangle the audible patterns of low-level prominence from the influence of higher-level prosody and substrate phonology. It is thus a fortunate and promising fact that the literature has recently begun to make rapid advances in each of these secondary arenas. The study of prosodic phrasing has begun to be taken up in many languages of the region, including especially Waima’a (spoken in East Timor; Himmelmann 2010) and Mandar (South Sulawesi; Brodtkin To Appear); fine interactions between foot structure and the edges of the  $\phi$  and  $\iota$  have begun to emerge in Mandar and other languages of South Sulawesi (Brodtkin, 2025a,b), and the link between word- and phrase-level prominence has begun to be studied in great detail in Betawi Malay by Van Heuven *et al.* 2008, Kupang Malay by Lai *et al.* 2010, Ambon Malay by Maskikit-Essed & Gussenhoven 2016, and Papuan Malay by Kaland 2019, 2021. For valuable review of the central issues, see Himmelmann & Kaufman 2018. As this body of work expands and integrates with corresponding research on prosodic phrasing and phrasal prominence in Standard Indonesian, it seems very likely that much will fall into place.

## References

- Abdullah, Hassan. 1974. *The Morphology of Malay*. Kuala Lumpur: Dewan Bahasa dan Pustaka.
- Adelaar, K Alexander. 1985. *Proto-Malayic: The reconstruction of its phonology and parts of its morphology and lexicon*. Ph.D. thesis, KanTERS.
- Adelaar, K Alexander. 2000. Malay: A short history. *Oriente moderno*, **19**(2), 225–242.
- Adisasmito, Niken. 1993. Syllable structure and the nature of schwa in Indonesian. *Studies in Linguistic Sciences*, **23**, 1–19.
- Alieva, Natalija Fedorovna, Arakin, Vladimir Dmitrievic, Ogloblin, Aleksandr Konstantinovic, & Sirk, Julo Chen-novic. 1972. *Grammatika indonezijskogo jazyka*. 'Nauka', Glavnaja redakcija vostochnoj literatury.
- Archangeli, Diana, Moll, Laura, & Ohno, Kazutoshi. 1998. Why not\* NC. vol. 34. Chicago: Chicago Linguistics Society.
- Badan Pusat Statistics Indonesia. 2020. *Hasil Long Form Sensus Penduduk 2020*.
- Blust, Robert. 2013. *The Austronesian Languages*. Canberra: Asia-Pacific Linguistics.
- Blust, Robert A. 2004. Austronesian nasal substitution: a survey. *Oceanic Linguistics*, **43**(1), 73–148.
- Brodkin, Dan. 2025a. *Reinforcers, Rightward Movement, and Prosodic Greed*. Resubmitted Manuscript.
- Brodkin, Dan. 2025b. Suppletion in Global Perspective. *Linguistic Inquiry*.
- Brodkin, Dan. To Appear. The Prosody of the Extended VP. *Natural Language and Linguistic Theory*.
- Cohn, Abigail, & McCarthy, John J. 1998. *Alignment and parallelism in Indonesian phonology*. Unpublished Manuscript: Cornell University, Ithaca, NY, and University of Massachusetts, Amherst.
- Cohn, Abigail C. 1989. Stress in Indonesian and bracketing paradoxes. *Natural language & linguistic theory*, **7**(2), 167–216.
- Court, Christopher. 1970. Nasal Harmony and some Indonesian Sound Laws. *Pages 203–218 of: Pacific Linguistic Studies in honour of Arthur Capell*. Canberra: The Australian National University.
- Dardjowidjojo, Soenjono. 1966. *Indonesian syntax*. Ph.D. thesis, Georgetown University.
- Delilkan, Ann S. 2002. *Fusion and other segmental processes in Malay: The crucial role of prosody*. Ph.D. thesis, New York University.
- Gil, David. 2020. What Does It Mean to Be an Isolating Language? The Case of Riau Indonesian. *Pages 9–96 of: Gil, D., & Schapper, A. (eds), Austronesian Undressed: How and Why Languages Become Isolating*. John Benjamins, Amsterdam.
- Gil, David. 2022. Monosyllables in Malay/Indonesian, a tale of two ludlings. *Markedness and non-concatenative morphology*, 45–100.
- Goedemans, Rob, & van Zanten, Ellen. 2007. Stress and accent in Indonesian. *LOT Occasional series*, **9**, 35–62.
- Grijns, CD. 1981. Jakartan speech and Takdir Alisjahbana's plea for the simple Indonesian word-form. *Pages 1–34 of: Wurm, S. A. (ed), Studies in Austronesian linguistics*. Ann Arbor: University of Michigan.
- Halim, Amran. 1974. *Intonation in Relation to Syntax in Indonesian*. Jakarta: Lembaga Bahasa Nasional & Penerbit Jambatan.



- Himmelman, Nikolaus, & Kaufman, Daniel. 2018. Some preliminary observations on prosody and information structure in Austronesian languages of Indonesia and East Timor. In: Riesberg, Sonja, Shiohara, Asako, & Utsumi, Atsuko (eds), *Studies in Diversity Linguistics*. Berlin: Language Science Press.
- Himmelman, Nikolaus P. 2010. Notes on Waima'a intonation. *East Nusantara: Typological and areal analyses*, 47–69.
- Inkelas, Sharon. 1989. *Prosodic constituency in the lexicon*. Ph.D. thesis, Stanford University.
- Itô, Junko, & Mester, Armin. 1992. *Weak layering and word binarity*. Manuscript; published 1992 as LRC-92-09, University of California, Santa Cruz.
- Ito, Junko, & Mester, Armin. 2007. Prosodic adjunction in Japanese compounds. *MIT working papers in linguistics*, 55, 97–111.
- Kaland, Constantijn. 2019. Acoustic correlates of word stress in Papuan Malay. *Journal of Phonetics*, 74, 55–74.
- Kaland, Constantijn. 2021. The perception of word stress cues in Papuan Malay: A typological perspective and experimental investigation. *Laboratory Phonology*, 12(1).
- Kurniawan, Ferdinan Okki. 2018. *Phonological variation in Jakarta Indonesian: An emerging variety of Indonesian*. Ph.D. thesis, Cornell University.
- Lai, Regine, Tynan, Emily, & Park, Yugyeong. 2010. *Stress Patterns in Kupangese*. Paper presented at ISMIL14, The Fourteenth International Symposium on Malay/Indonesian Linguistics, Minneapolis, MN, USA, 2 May.
- Lapoliwa, Hana. 1981. *A generative approach to the phonology of Bahasa Indonesia*. Pacific Linguistics D-36. Canberra: RSPAS.
- Mahdi, Waruno. 1981. Some problems of the phonology of metropolitan Indonesian. *Bijdragen tot de Taal-, Land-en Volkenkunde*, 399–418.
- Maskikit-Essed, Raechel, & Gussenhoven, Carlos. 2016. No stress, no pitch accent, no prosodic focus: The case of Ambonese Malay. *Phonology*, 33(2), 353–389.
- McCarthy, John J. 2004. *Headed spans and autosegmental spreading*. Unpublished ms.
- Nomoto, Hiroki. 2009. More on Austronesian nasal substitution. *Pages 503–517 of: Proceedings from the Annual Meeting of the Chicago Linguistic Society*, vol. 45. Chicago: Chicago Linguistics Society.
- Pater, Joe. 1996. \*NC. GLSA, University of Massachusetts, Amherst.
- Pater, Joe. 1999. Austronesian nasal substitution and other \*NC effects. *Pages 310–343 of: Kager, René, van der Hulst, Harry, & Zonneveld, Wim (eds), The Prosody-Morphology Interface*. Cambridge: Cambridge University Press.
- Pater, Joe. 2001. Austronesian nasal substitution revisited. 159–182.
- Robins, Robert H. 1953. The phonology of the nasalized verbal forms in Sundanese. *Bulletin of the School of Oriental and African Studies*, 15(1), 138–145.
- Samsuri. 1960. Sistem fonem Indonesia dan Suatu Penjusunan Edjaan Baru. *Medan: Ilmu Pengetahuan*, 1(4), 323–341.
- Selkirk, Elisabeth. 2009. On clause and intonational phrase in Japanese: The syntactic grounding of prosodic constituent structure. *Gengo Kenkyu*, 136, 35–73.
- Sneddon, James Neil. 1996. *Indonesian: A comprehensive grammar*. Routledge.
- Soh, Hooi Ling. 2011. Phonological opacity in Malay: Its relevance for constraint evaluations in the Optimality Theory. In: *Isamu Shoho: Tinta Kenangan "Kumpulan Esei Bahasa dan Linguisti"*. Kuala Lumpur: Dewan Bahasa dan Pustaka.

- Stokhof, WAL. 1975. On the phonology of Bahasa Indonesia. *Bijdragen tot de Taal-, Land-en Volkenkunde*, 254–269.
- Tadmor, Uri. 2007. Grammatical borrowing in Indonesian. *Page 301 of: Matras, Yaron, & Sakel, Jeanette (eds), Empirical Approaches To Language Typology*, vol. 38. Mouton de Gruyter.
- Teeuw, Andries. 1984. *Leerboek Bahasa Indonesia (Indonesian Textbook)*. Groningen: WoltersNoordhoff (4th ed).
- Van Heuven, Vincent J, Roosman, Lilie, & Van Zanten, Ellen. 2008. Betawi Malay word prosody. *Lingua*, **118**(9), 1271–1287.
- van Zanten, Ellen, & van Heuven, Vincent J. 1998. Word stress in Indonesian: Its communicative relevance. *Bijdragen tot de Taal-, Land-en Volkenkunde*, **154**(1), 129–149.
- Walker, Rachel. 1998. *Nasalization, neutral segments and opacity effects*. Ph.D. thesis, University of California, Santa Cruz.
- Wallace, Stephen. 1976. *Linguistic and Social Dimensions of Phonological Variation in Jakarta Malay*. Ph.D. thesis, Cornell University.
- Wedhawati, Erni Siti Nurlina, Wiwin, Laginem, Nardiati, Sri, Herawati, Sukesti, Restu, Marsono, Setiyanto, Edi, Sabariyanto, Dirgo, Arifin, Syamsul, & Sumadi. 2001. *Tata Bahasa Jawa Mutakhir (A Contemporary Grammar of Javanese)*. Jakarta: Badan Pengembangan dan Pembinaan Bahasa.